FROM THE
EDITOR-IN-CHIEF

For a country on the vanguard of the software revolution, India has curiously stayed a few light years behind the hardware architecture that supports it: semiconductor chips. India imports 95 per cent of its annual requirement of 19 billion chips, paying a hefty import bill of Rs 1.71 lakh crore. Why do chips matter? Because this tiny silicon wafer, with billions of nano-sized transistors embossed on a base of just a few millimetres, is the beating heart of the Information Age. Almost everything runs on this complex little marvel: from your slim mobile phone to data centres spread across millions of square feet, from automated factories to the robots that do the grunt work, from automatic cars to every other navigable object like ships, aircraft, rockets and spacecraft, from your smart TV to your WiFi router, from MRI scanners to cloud services. More recently, Operation Sindoor demonstrated how central a role electronic gadgetry plays in modern warfare. Whether through autonomous intelligent drones, precision-guided missiles or radar tracking and surveillance systems, it's information transmitted across chips that play a decisive role in the outcome. No wonder the silicon chip is seen as the steel of the Fourth Industrial Revolution: it forms the core of just about everything.

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Trouble is, India does not make it. The folly of that approach came out starkly during the COVID-19 pandemic, when supply chain disruptions created a massive semiconductor crisis globally and impacted several key industries in India, including the auto and consumer electronic goods sectors. Even a country like the US suffered. Though it leads in chip design and innovation, it presently accounts for only 12 per cent of the world's chip manufacturing. The fabrication industry is dominated by East Asia: Taiwan, South Korea and, to some extent, China. That's what goaded America, under President Joe Biden in 2022, to pass the Chips and Science Act: it set aside $52.7 billion to boost domestic semiconductor manufacturing. India had already decided that it too could not afford such upheavals and needed to be self-reliant in chips. Under the India Semiconductor Mission launched in December 2021, the Modi government made one of its biggest gambits. It pledged to invest Rs 76,000 crore to attract private chip manufacturers, both foreign and Indian, to set up foundries here.

Unlike the Performance Linked Incentive (PLI) scheme, it offered to underwrite up to half the project cost but with no repayment or equivalent equity, simply to seed a fully homegrown ecosystem of chip manufacturing. This would span across designing and fabricating chips to having them assembled, tested and packaged indigenously. States were also encouraged to provide grants ranging from 20 to 30 per cent of the project costs.

Chips are not easy to make and only a handful of nations do it. That's because it needs ultra-precision, up to 99.5 per cent accuracy, to create a tiny silicon wafer treated with photosensitive agents, transfer the chip circuit design on to it using lithography, and fire billions of transistors and the rest of the circuitry on to it. It's capital-intensive and creates few jobs. And besides high order engineering skills, it needs enormous supplies of uninterrupted power, water and rare chemicals. Such massive investments by the Modi government drew criticism from Raghuram Rajan, the former Reserve Bank of India governor, who described India's mission as a "ruinous race to get into now", pointing to its high capital costs, heavy upfront subsidies and low labour potential. Ashwini Vaishnaw, Union Minister for Electronics and Information Technology, brushes aside the criticism. "It is a foundational industry that is of strategic importance to India, especially in a world full of turbulence and geopolitical changes. We need to be self-reliant," he tells INDIA TODAY in an interview. "It is a very complex industry compared to any other manufacturing industry and must be supported by government. That is the practice the world over."

Powered by cumulative investments of around Rs 1.6 lakh crore, it's a bold, risky and expensive plan, with a gestation period of about a decade. But the bet seems to be already paying off. By now, 10 projects have been greenlit and fast-tracked into construction mode. From big players like Tata Electronics, partnering with the Taiwanese giant PSMC for India's first semiconductor fabrication unit, to US major Micron Technology and an HCL-Foxconn venture, the Indian landscape is now bristling with activity. Four smaller units were set rolling just in August, and startups are foraying into advanced chip design. Group Editorial Director Raj Chengappa, who has tracked the frontiers of science and technology for a long time, brings all his expertise into this week's cover story.

Chengappa visited Sanand in Gujarat, one of the new semiconductor hubs, for a close-up look at Murugappa Group's CG Semi facility as it became the first to launch its pilot line. He was impressed with the cutting-edge technology employed, apart from the high-quality standards. A space and atomic energy buff, Chengappa says, "The technology to make chips is confined to an exclusive club of nations and it has become imperative for India to become a full-fledged member."

It's a blue-chip dream that needs to become a reality. So that when the chips are down, our chips are up. Fingers firmly crossed.

(Aroon Purie)